

Question				Marking details			Marks Available					
							AO1	AO2	AO3	Total	Maths	Prac
10	(a)	(i)			Sympathetic	Parasympathetic	3			3		
				Neurotransmitter	Noradrenaline NOT adrenaline	Acetylcholine (1)						
				General effect	Excitatory/ stimulatory;	Inhibitory (1)						
				Effect on heart rate	Increases	Decreases; (1)						
				1 mark for each row								
		(ii)		The larger the area on the motor cortex + increased movement {complexity/ increased fine movements} NOT number of motor neurones			1			1		
		(iii)		Increased sensory cortex to areas more receptive to stimuli / or example increased sensory cortex for genitals/ hand/ tongue etc (1)				1		1		
	(b)	(i)		Cortisol levels raised (1) Cortisol levels raised all the time (1) Negative Feedback loop {not working/ overridden/ faulty} (1)				2	1	3		
		(ii)		Epigenetic factors (1) Increased gene expression leading to increased cortisol levels (1)				2		2		
	(c)	(i)		Division of labour/ separate working groups/ different groups for different jobs/ 2 examples from: care of offspring/ looking for food/ reproduction/ defence of the colony (1) Increased efficiency (1)				2		2		
		(ii)		Pheromones/visual cues / touch/ movement				1		1		
		(iii)		Intra-sexual (selection)(1) Males more aggressive / increases sexual dimorphism / weaker males die off / strong males {survive/ reproduction}/ passing on of advantageous alleles (1)			1	1		2		